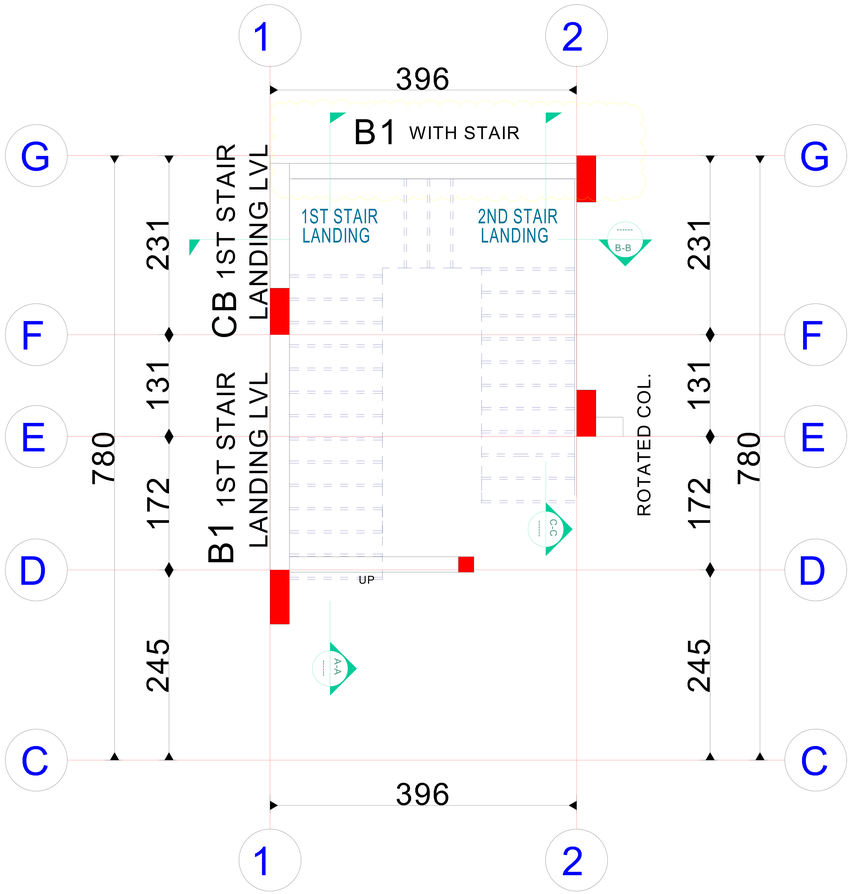
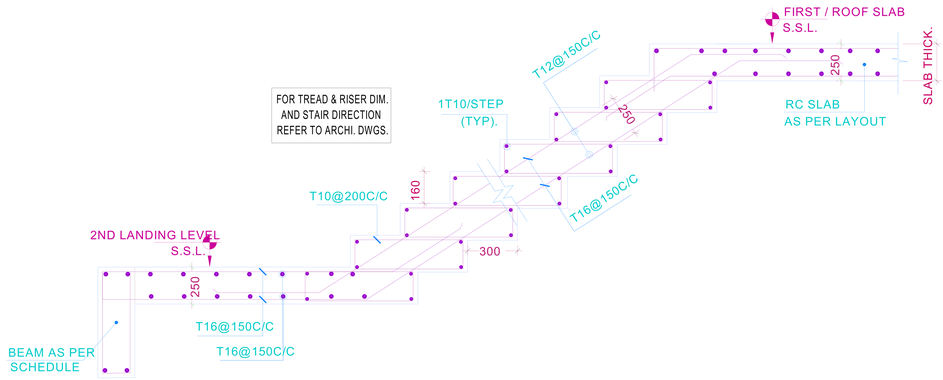
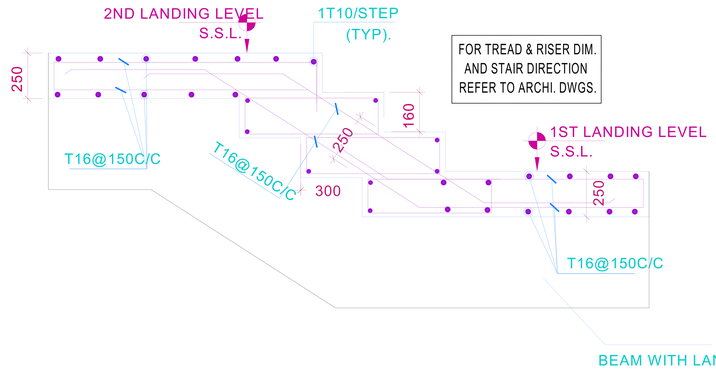




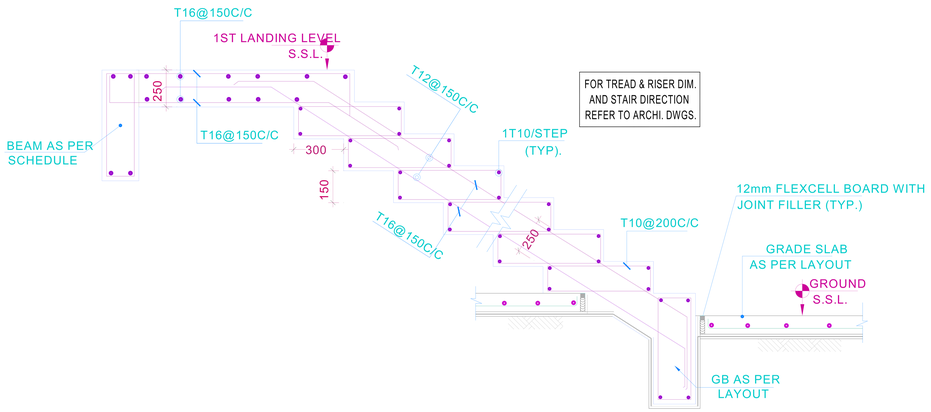
SECTION FROM 2ND LANDING LEVEL TO FIRST / ROOF FLOOR LEVEL

SEC(C-C)



SECTION FROM 1ST LANDING LEVEL TO 2ND LANDING LEVEL

SEC(B-B)



SECTION FROM GROUND FLOOR TO 1ST LANDING LEVEL

SEC(A-A)

STAIR DETAILS

Govemment Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

حكومة رأس الخيمة
دائرة البلدية
إدارة المباني

Approved On — اعتمد في

15 November 2024

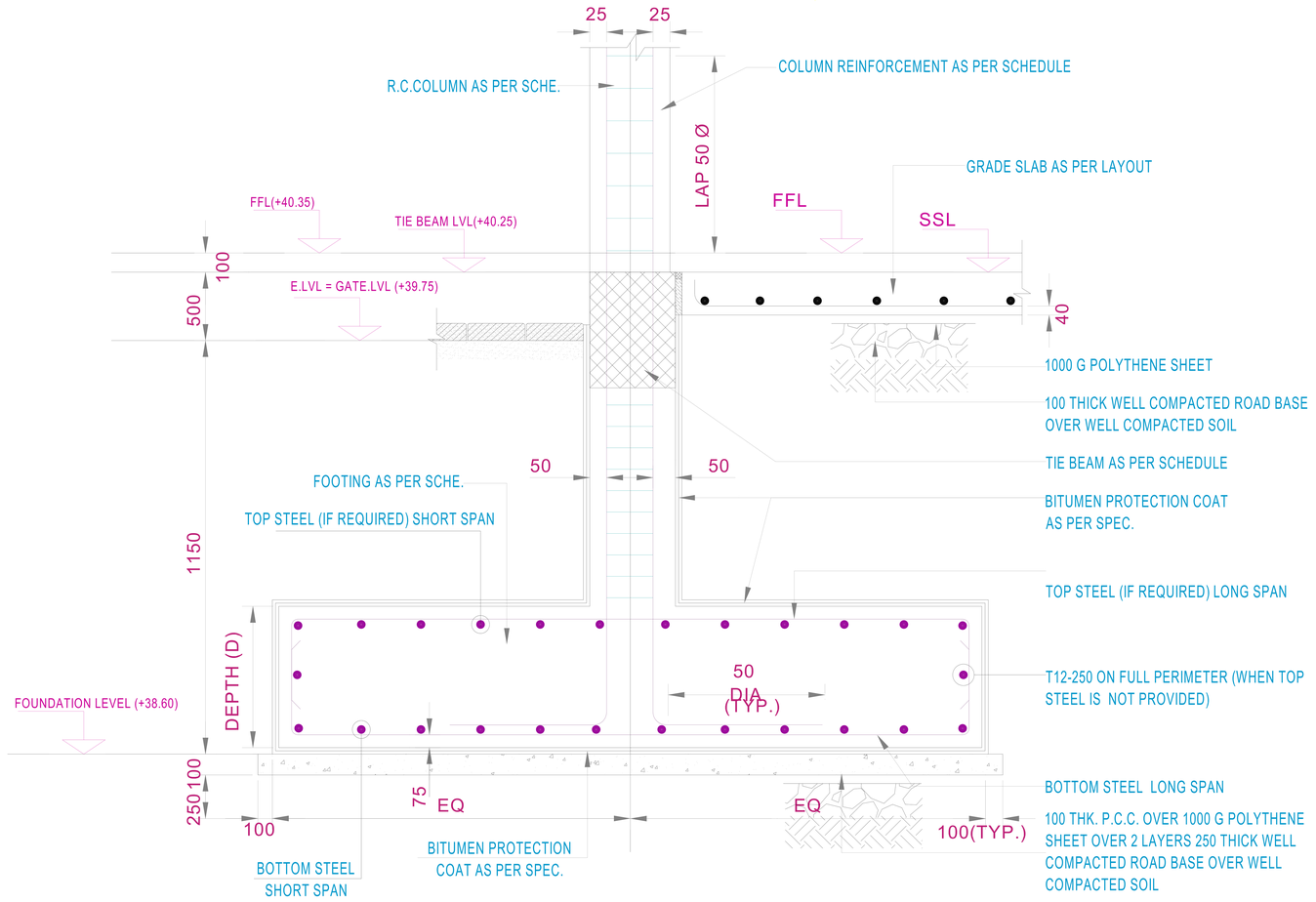
Building Permit #: PRMT-24-0005227

Approved By:

Savana
Engineering Consultancy & interior design

Client	SAEED AHMED MOHAMMED HASAN ALMANSOORI		
Project	DOUBLE STORY VILAA		
Drg.Title	STAIR DETAILS		
Area	ARIFAA	Floortype	
Case ID.	2673356	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	607011962
Bldg. No.	BLD-24-03157	Floor No.	
Date	OCT-2024	Sht. No.	S07
Scale	1/100	Job No.	

G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG.IBRAHIM



(TYPICAL SECTION FOR FOOTING)

ALLOWABLE BEARING CAPACITY FOR ISOLATED/ STRIP FOOTINGS		
Width of Footing	Allowable Bearing Pressure	Modulus of sub-grade
Up to 2.00m	140 kN/m2	16,800 kN/m3
Up to 3.00m	130 kN/m2	15,600 kN/m3
Up to 4.00m	120 kN/m2	14,400 kN/m3

- Excavate and level the existing soil at the Level of -1.50m below the E.LVL established on the ground level of the existing in the site .

- Place ONE (1) layer of selected granular backfill materials (Road Base) 25cm thick to reach the required foundation level. the layer shall be compacted by heavy vibratory roller to a degree of compaction not less than 95% of the maximum dry density as obtained by modified proctor compaction test (ASTM D 1557) .

- A 100mm thick layer of blinding concrete (PCC) should be placed to protect the compacted surface .

- Shallow foundation can be placed on engineered fill. (i.e., Approximately -1.15m below the E.LVL established on the ground level of the existing in the site) .

- Plate bearing test must be carried out at foundation level as quality control measure to verify the required allowable bearing pressure and total settlement criteria under foundations .

DETAILS FONDATION

Govemment Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration
حكومة رأس الخيمة
دائرة البلدية
إدارة المباني

Approved On / اعتمد في

15 November 2024

Building Permit #: PRMT-24-0005227

Appoved By: Reham Hussein

STRUCTURAL NOTES

ACI 318-11
F'C/FCU = 28/35 N/mm2
FY = 460N/mm2
FYt = 420N/mm2

CONCRETE COVERS :

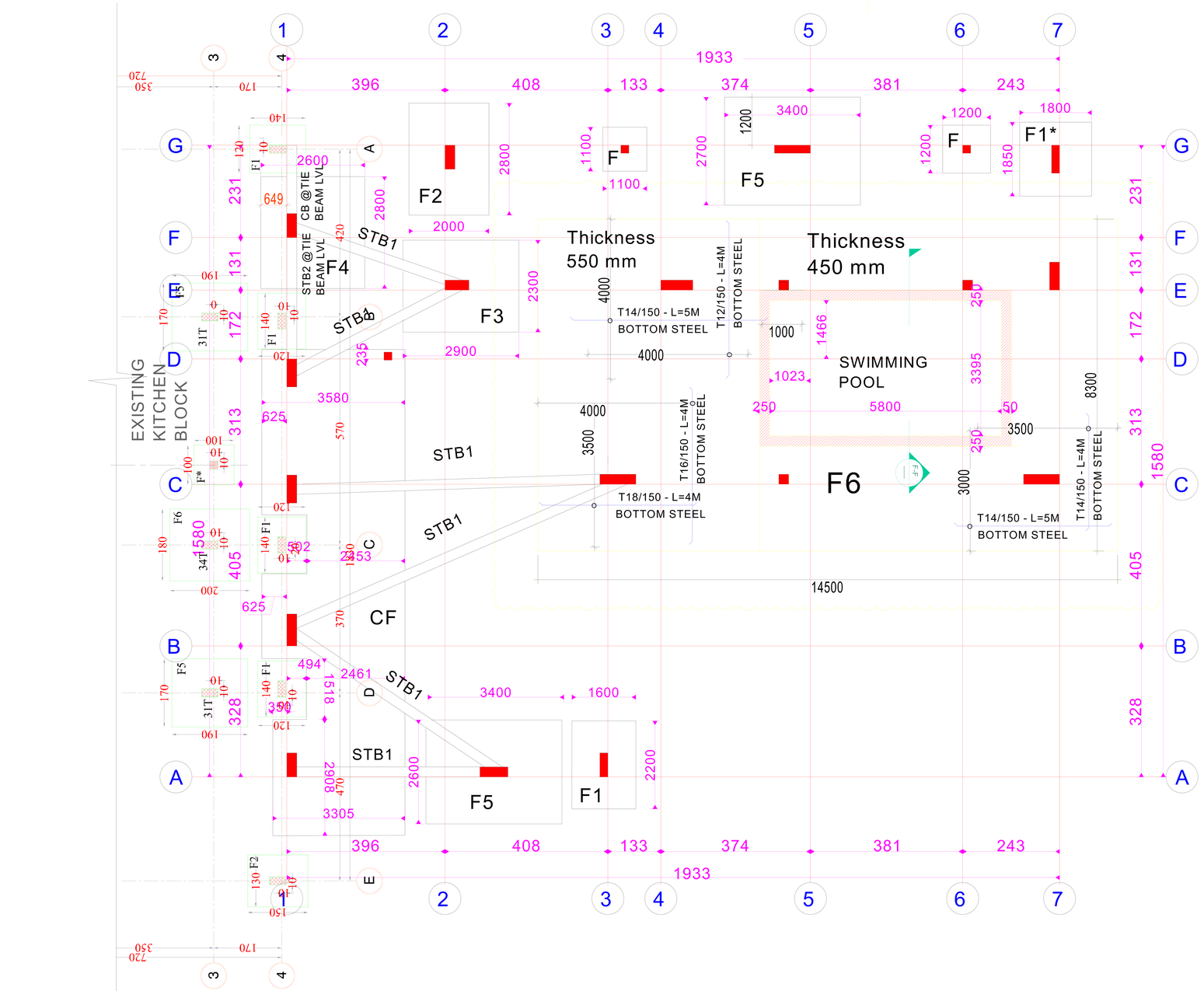
Foundations cover = 75mm
Columns cover = 40mm
Walls cover = 30mm
Beams cover = 40mm
Slabs & Stairs cover = 25mm

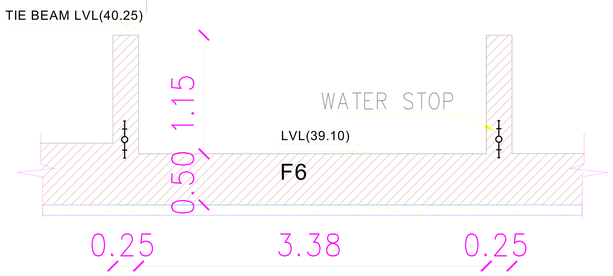
Rooms	Space	Number
Residential		
Commercial		
Industrial		
Goverment		
Resident.-Comm.		
Total		

Savana
Engineering Consultancy & Interior design

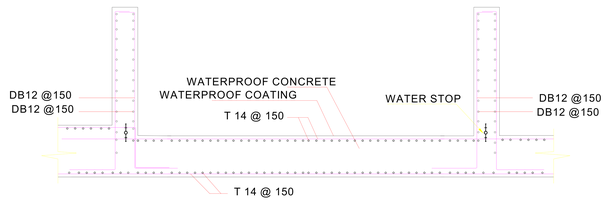
Client	SAEED AHMED MOHAMMED HASAN ALMANSOORI		
Project	DOUBLE STORY VILAA		
Drg.Title	DETAILS FOUNDATION		
Area	ARIFAA	Floortype	
Case ID.	2673356	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	607011962
Bldg. No.	BLD-24-03157	Floor No.	
Date	OCT-2024	Sht. No.	S01
Scale	1/100	Job No.	

G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG.IBRAHIM





section (f-f)
CONCRETE DETAIL



section (f-f)
STEEL DETAIL

FONDATION LAYOUT

Govement Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

حكومة رأس الخيمة
دائرة البلدية
إدارة المباني

Approved On — اعتمد في

15 November 2024

Building Permit #: PRMT-24-0005227

Approved By:

STRUCTURAL NOTES

ACI 318-11

F'C/FCU = 28/35 N/mm2

FY = 460N/mm2

FYt = 420N/mm2

CONCRETE COVERS :

Foundations cover = 75mm

Columns cover = 40mm

Walls cover = 30mm

Beams cover = 40mm

Slabs & Stairs cover = 25mm

Rooms	Space	Number
Residential		
Commercial		
Industrial		
Goverment		
Resident.-Comm.		
Total		

Savana
Engineering Consultancy & Interior design

Client	SAEED AHMED MOHAMMED HASAN ALMANSOORI		
Project	DOUBLE STORY VILAA		
Drg.Title	FOUNDATION LAYOUT		
Area	ARIFAA	Floortype	
Case ID.	2673356	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	607011962
Bldg. No.	BLD-24-03157	Floor No.	
Date	OCT-2024	Sht. No.	S02
Scale	1/100	Job No.	

G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG.IBRAHIM

TYPE	SIZE			REINFORCEMENT DETAILS				REMARKS.
				BOTTOM MESH		TOP MESH		
	WIDTH (mm)	LENGTH (mm)	DEPTH (mm)	SHORT BAR	LONG BAR	SHORT BAR	LONG BAR	
F	1200	1200	400	T12-150	T12-150	4T12	4T12	
F*	1000	1400	400	T14-125	T14-125	4T12	4T12	
F1	1600	2200	400	T14-150	T14-150	4T12	4T12	
F1*	1800	1850	400	T14-150	T14-150	4T12	4T12	
F2	2000	2800	400	T14-150	T14-150	4T12	4T12	
F3	2300	2900	450	T14-150	T14-150	4T12	4T12	
F4	2600	2800	400	T14-150	T14-150	4T14	4T14	
F5	2700	3400	450	T16-150	T16-150	4T12	4T12	
F6	8300	14500	450&550	T14-150	T14-150	T14-150	T14-150	SWIMMING POOL FOOTING
CF	AS PER LAYOUT	12150	400	T12-125	T12-125	T12-125	T12-125	

SCHEDULE OF STB

TYPE	BEAM SIZE	BOTTOM REINFORCEMENT		TOP REINFORCEMENT			SIDE BARS O.E.F (F) EACH SIDE	STIRRUPS				NO OF LEGS	REMARKS
								AT SUPPORT		AT MIDSPAN			
	MM X MM	STRAIGHT (A)	EXTRA @ MID SPAN(B)	STRAIGHT (C)	EXTRA @ CONT. SUPPORT (D)	EXTRA @ DISCONT. SUPPORT (E)		OUTER (G1)	INNER (G2)	OUTER (H1)	INNER (H2)		
STB1	250 x1300	4T18	--------	3T18+3T18	--------	--------	--------	T10-100	--------	T10-150	--------	2	
STB2	250 x1200	3T18	--------	2T18+2T18	--------	--------	--------	T10-100	--------	T10-150	--------	2	AT TIE BEAM LEVEL
STB3	200 x1300	2T18+2T18	--------	3T18+3T18	--------	--------	--------	T10-100	--------	T10-150	--------	2	

SCHEDUEL FONDATION

Govemment Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

حكومة رأس الخيمة
دائرة البلدية
إدارة المباني

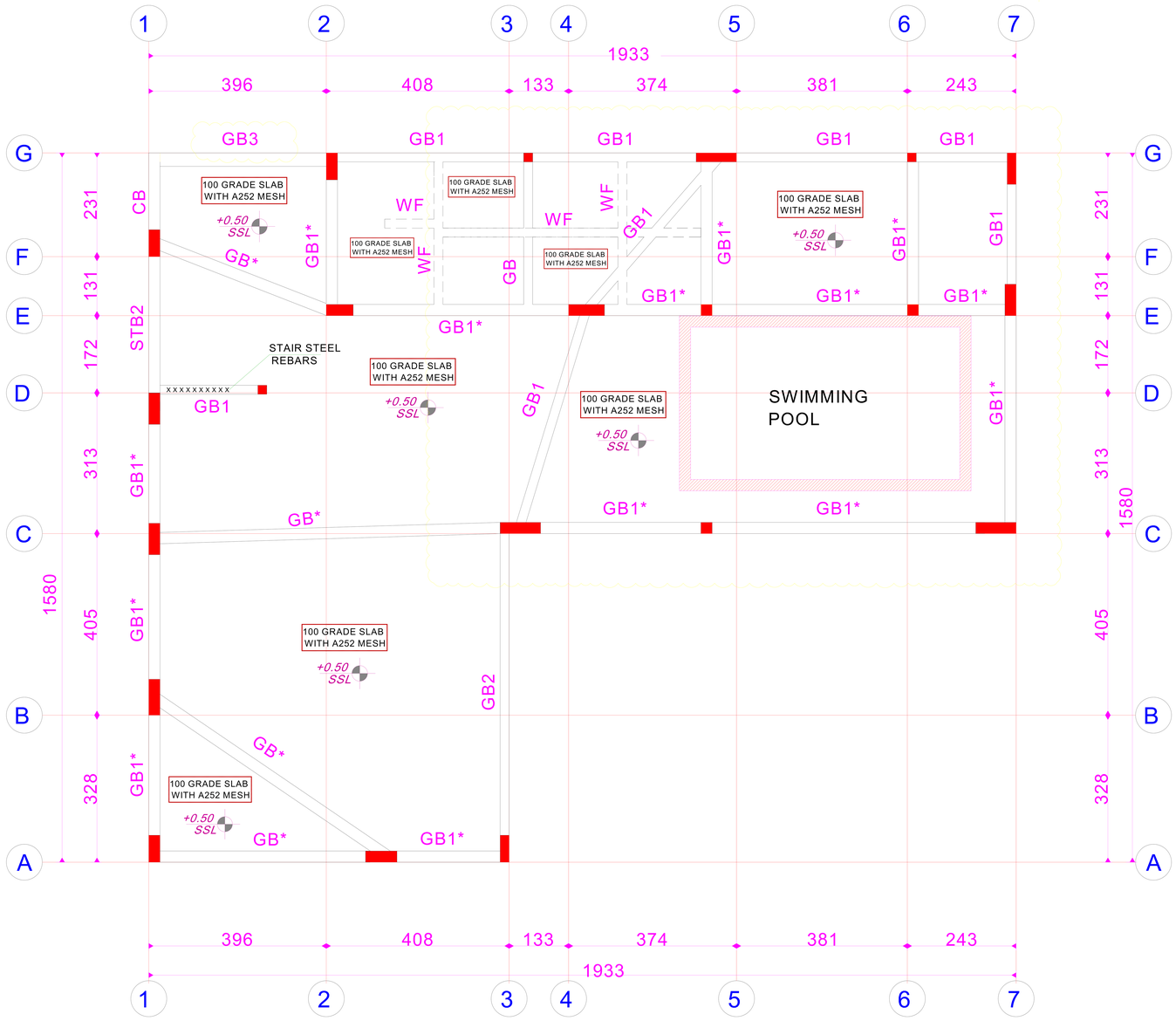
Approved On: 15 November 2024
اعتمد في

Building Permit #: PRMT-24-0005227
Appoved By: Ibham Hussein

Savana
Engineering Consultancy & interior design

Client	SAEED AHMED MOHAMMED HASAN ALMANSOORI		
Project	DOUBLE STORY VILAA		
Drg.Title	FOUNDATION SCHEDULE		
Area	ARIFAA	Floortype	
Case ID.	2673356	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	607011962
Bldg. No.	BLD-24-03157	Floor No.	
Date	OCT-2024	Sht. No.	S03
Scale	1/100	Job No.	

G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG.IBRAHIM



SCHEDULE OF GROUND BEAMS

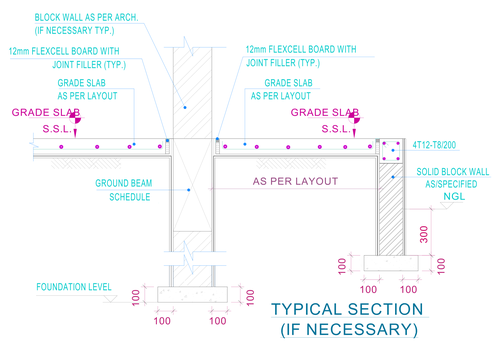
TYPE	BEAM SIZE MM XMM	STIRRUPS AT SUPPORT OUTER (G1)	STIRRUPS AT SUPPORT INNER (G2)	STIRRUPS AT MIDSPAN OUTER (H1)	STIRRUPS AT MIDSPAN INNER (H2)	NO OF LEGS	REMARKS
GB*	250 X 350	T8-150	------	T8-200	------	2	ABOVE STB
GB	200 X 500	T8-150	------	T8-200	------	2	
GB1	200 X 600	T8-150	------	T8-200	------	2	
GB1*	250 X 600	T8-150	------	T8-200	------	2	
GB2	200 X 700	T10-150	------	T10-200	------	2	
GB2*	250 X 700	T10-150	------	T10-200	------	2	
GB3	300 X 600	T10-150	------	T10-200	------	2	

TYPE	SIDE BARS O.E.F (F) EACH SIDE	TOP REINFORCEMENT EXTRA @ DISCONT. SUPPORT (E)	TOP REINFORCEMENT EXTRA @ CONT. SUPPORT (D)	TOP REINFORCEMENT STRAIGHT (C)	BOTTOM REINFORCEMENT STRAIGHT (A)	BOTTOM REINFORCEMENT EXTRA @ MID SPAN(B)
GB*	1T12 : :	----	----	2T16	2T16	----
GB	1T12 : :	----	----	2T16	2T16	----
GB1	1T12 : :	1T16	1T16	2T16	2T16	1T16
GB1*	1T12 : :	1T16	1T16	2T16	3T16	2T14
GB2	2T12 : :	2T14	2T14	2T16	3T16	2T14
GB2*	2T12 : :	2T14	2T14	2T16	3T16	2T14
GB3	2T12 : :	2T14	2T14	2T16	3T16	2T14

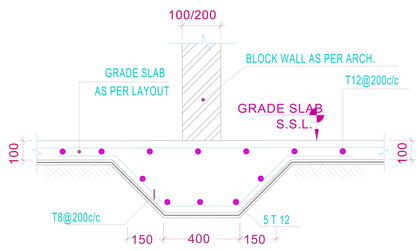
SCHEDULE OF STB

TYPE	BEAM SIZE MM X MM	STIRRUPS AT SUPPORT OUTER (G1)	STIRRUPS AT SUPPORT INNER (G2)	STIRRUPS AT MIDSPAN OUTER (H1)	STIRRUPS AT MIDSPAN INNER (H2)	NO OF LEGS	REMARKS
STB2	250 x1200	T10-100	-------	T10-150	-------	2	AT TIE BEAM LEVEL
CB	250 x1200	T10-100	-------	T10-150	-------	2	AT TIE BEAM LEVEL

TYPE	BEAM SIZE MM X MM	SIDE BARS O.E.F (F) EACH SIDE	TOP REINFORCEMENT EXTRA @ DISCONT. SUPPORT (E)	TOP REINFORCEMENT EXTRA @ CONT. SUPPORT (D)	TOP REINFORCEMENT STRAIGHT (C)	BOTTOM REINFORCEMENT STRAIGHT (A)	BOTTOM REINFORCEMENT EXTRA @ MID SPAN(B)
STB2	250 x1200	4T12 : :	-------	-------	2T18+2T18	3T18	--------
CB	250 x1200	4T12 : :	-------	-------	3T18+3T18	3T18	--------

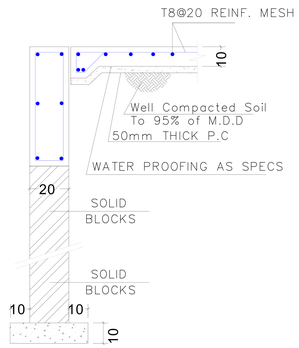


TYPICAL SECTION (IF NECESSARY)

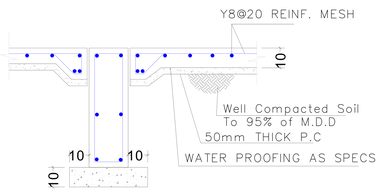


TYPICAL DETAILS FOR (W.F.)
FOOTING FOR 10cm/20cm LOAD BEARING WALLS.

TYPICAL DETAILS



TYPICAL EXTERNAL GROUND BEAM
SC. (1/25)



TYPICAL INTERNAL GROUND BEAM
SC. (1/25)

TIE BEAM LAYOUT

Govemment Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

حكومة رأس الخيمة
دائرة البلدية
إدارة المباني

Approved On — اعتمد في

15 November 2024

Building Permit #: PRMT-24-0005227

Approved By:

STRUCTURAL NOTES

ACI 318-11
F'C/FCU = 28/35 N/mm2
FY = 460N/mm2
FYt = 420N/mm2

CONCRETE COVERS :

Foundations cover = 75mm
Columns cover = 40mm
Walls cover = 30mm
Beams cover = 40mm
Slabs & Stairs cover = 25mm

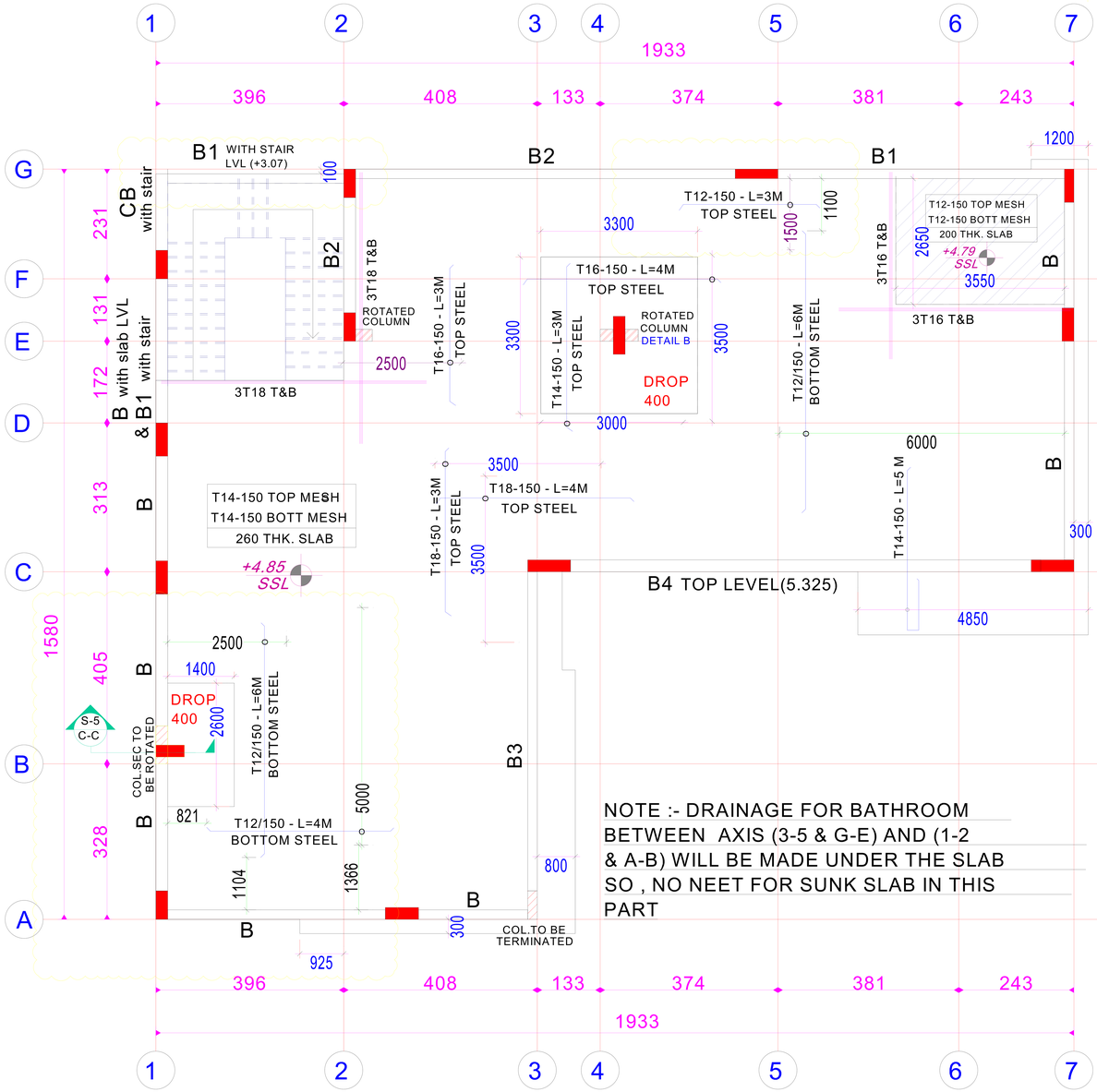
LEGEND:

1) COLUMN / WALL TO BE CONTINUED
2) COLUMN / WALL TO BE TERMINATED

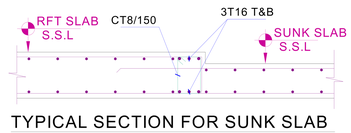
Savana
Engineering Consultancy & interior design

Client	SAEED AHMED MOHAMMED HASAN ALMANSOORI		
Project	DOUBLE STORY VILAA		
Drg.Title	TIE BEAM LAYOUT		
Area	ARIFAA	Floortype	
Case ID.	2673356	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	607011962
Bldg. No.	BLD-24-03157	Floor No.	
Date	OCT-2024	Sht. No.	S04
Scale	1/100	Job No.	

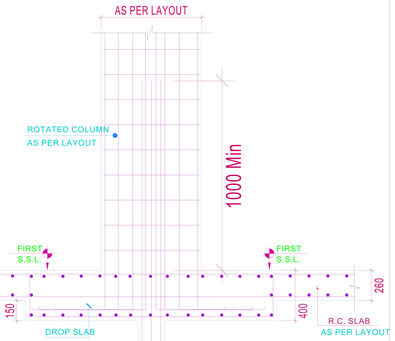
G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG.IBRAHIM



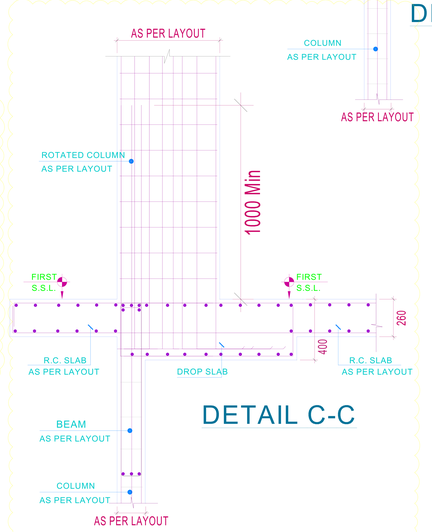
NOTE :- DRAINAGE FOR BATHROOM BETWEEN AXIS (3-5 & G-E) AND (1-2 & A-B) WILL BE MADE UNDER THE SLAB SO , NO NEET FOR SUNK SLAB IN THIS PART



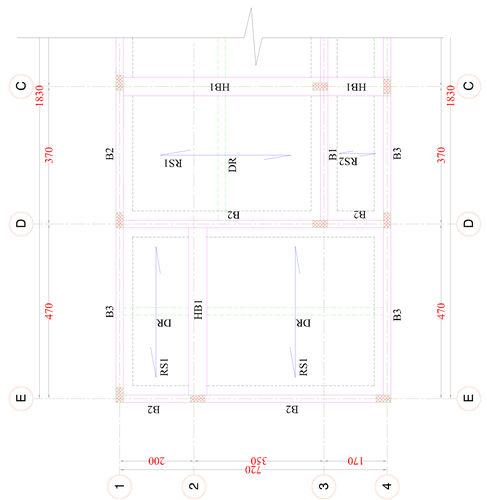
TYPICAL SECTION FOR SUNK SLAB



DETAIL B



DETAIL C-C



NOTE :- ACC BLOCKS WILL BE USED UBOVE KITCHEN BLOCK SLAB

SCHEDULE OF FIRST BEAMS

TYPE	BEAM SIZE	BOTTOM REINFORCEMENT		TOP REINFINFORCEMENT			SIDE BARS (F)	STIRRUPS					
	MM XMM	STRAIGHT (A)	EXTRA @ MID SPAN(B)	STRAIGHT (C)	EXTRA @ CONT. SUPPORT (D)	EXTRA @ DISCONT. SUPPORT (E)	EACH SIDE	AT SUPPORT OUTER (G1)	AT SUPPORT INNER (G2)	AT MIDSPAN OUTER (H1)	AT MIDSPAN INNER (H2)	NO OF LEGS	REMARKS
B	200X700	3T16	2T14	2T16	1T14	1T14	2T12	T10-200	------	T10-200	------	2	
B1	200X900	3T16	2T16	3T16	2T16	2T16	3T12	T10-150	------	T10-200	------	2	
B2	200X1000	3T16	3T16	3T16	2T16	2T16	3T12	T10-150	------	T10-200	------	2	
B3	200X970	3T16	3T16 + 2T16	3T16	3T16	3T16	4T12	T10-150	------	T10-200	------	2	
B4	250X1450	2T20+2T20	2T20	3T16	2T16	2T16	4T12	T10-150	------	T10-200	------	2	
CB	200X900	3T16	------	3T16+3T16			3T12	T10-150	------	T10-200	------	2	

Govement Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

حكومة رأس الخيمة
دائرة البلدية
إدارة المباني

Approved On — اعتمد في

15 November 2024

Building Permit #: PRMT-24-0005227

Appoved By:

LEGEND:

- COLUMN / WALL TO BE CONTINUED
- COLUMN / WALL TO BE TERMINATED

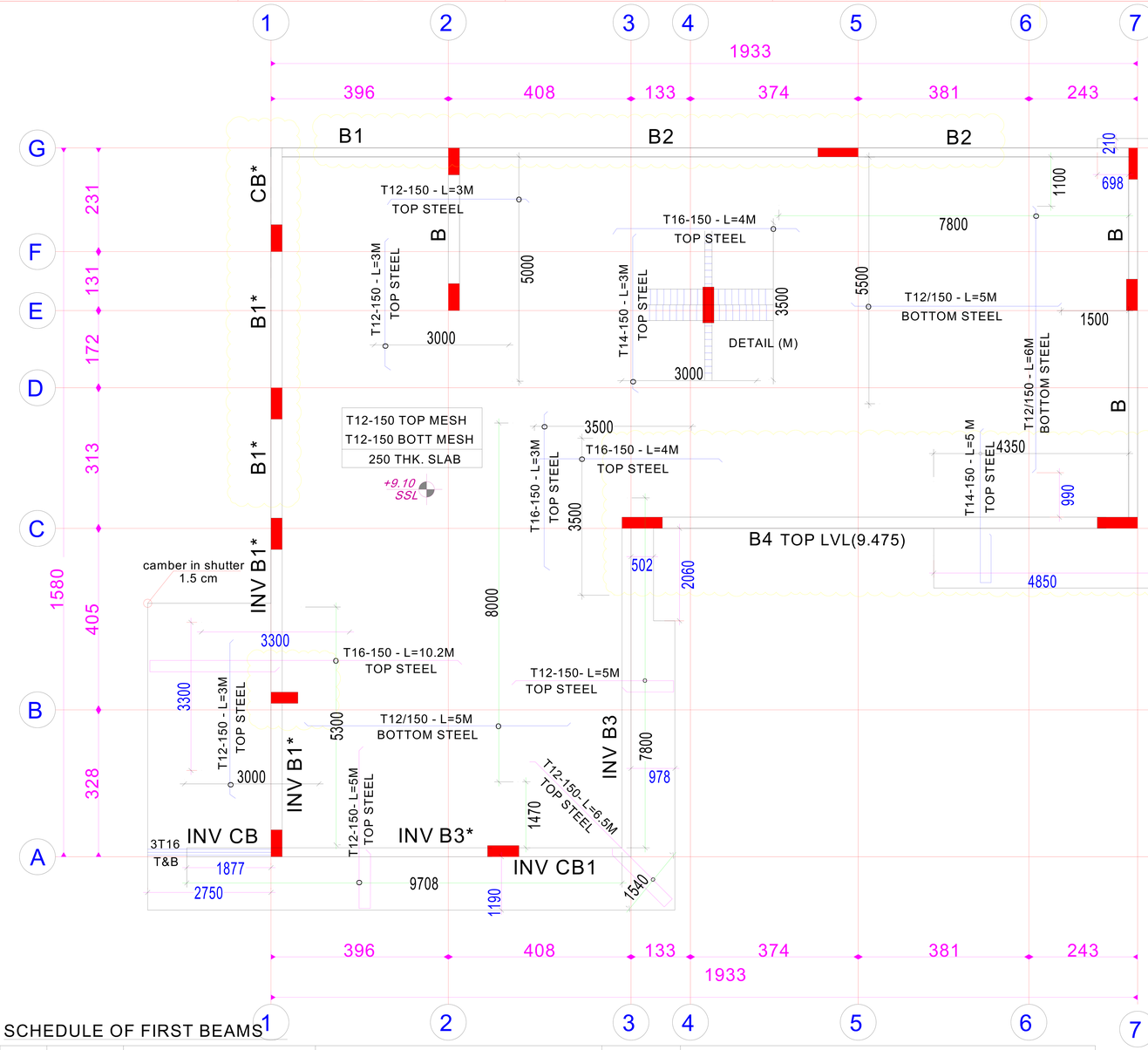
STRUCTURAL NOTES	CONCRETE COVERS :
ACI 318-11	Foundations cover = 75mm
F'C/FCU = 28/35 N/mm2	Columns cover = 40mm
FY = 460N/mm2	Walls cover = 30mm
FYt = 420N/mm2	Beams cover = 40mm
	Slabs & Stairs cover = 25mm

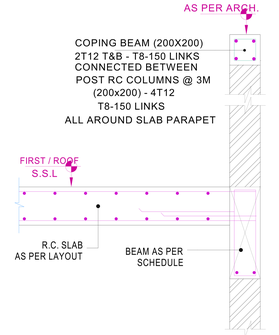
Savana
Engineering Consultancy & Interior design

Client	SAEED AHMED MOHAMMED HASAN ALMANSOORI		
Project	DOUBLE STORY VILAA		
Drg.Title	GROUND FLOOR SLAB-BEAMS		
Area	ARIFAA	Floortype	
Case ID.	2673356	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	607011962
Bldg. No.	BLD-24-03157	Floor No.	
Date	OCT-2024	Sht. No.	S05
Scale	1/100	Job No.	

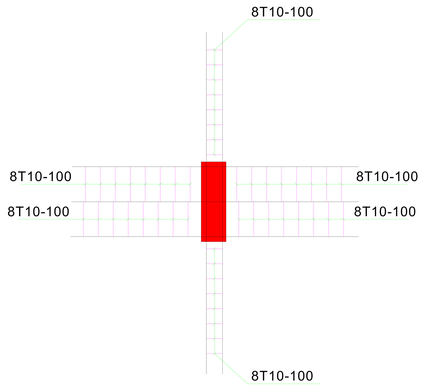
G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG.IBRAHIM

GROUND FLOOR SLAB-BEAMS





TYPICAL SECTION FOR PARAPET



DETAIL (M)

LEGEND:

1) COLUMN / WALL TO BE CONTINUED

2) COLUMN / WALL TO BE TERMINATED

STRUCTURAL NOTES	CONCRETE COVERS :
ACI 318-11	Foundations cover = 75mm
F'C/FCU = 28/35 N/mm2	Columns cover = 40mm
FY = 460N/mm2	Walls cover = 30mm
FYt = 420N/mm2	Beams cover = 40mm
	Slabs & Stairs cover = 25mm

SCHEDULE OF FIRST BEAMS

TYPE	BEAM SIZE MM XMM	BOTTOM REINFORCEMENT STRAIGHT (A)	EXTRA @ MID SPAN(B)	TOP REINFINFORCEMENT STRAIGHT (C)	EXTRA @ CONT. SUPPORT (D)	EXTRA @ DISCONT. SUPPORT (E)	SIDE BARS (F) EACH SIDE	STIRRUPS AT SUPPORT OUTER (G1)	AT SUPPORT INNER (G2)	AT MIDSPAN OUTER (H1)	AT MIDSPAN INNER (H2)	NO OF LEGS	REMARKS
B	200X700	3T16	2T14	2T16	1T14	1T14	2T12	T10-200	------	T10-200	------	2	
B1*	250X800	3T16	2T16	2T16	2T14	2T14	3T12	T10-150	------	T10-200	------	2	
B1	200X800	3T16	2T16	2T16	2T14	2T14	3T12	T10-150	------	T10-200	------	2	
B2	200X1000	3T16	3T16	2T16	2T16	2T16	3T12	T10-150	------	T10-200	------	2	
B3	200X1200	3T16	3T16 + 2T16	2T16	2T16	2T16	4T12	T10-150	------	T10-200	------	2	
B3*	200X1200	3T16	3T16	3T20	3T20	3T20	4T12	T10-150	------	T10-200	------	2	
B4	250X1200	2T20+2T20	2T20	3T16	2T16	2T16	4T12	T10-150	------	T10-200	------	2	
CB*	200X800	3T16	------	3T16+3T16			3T12	T10-150	------	T10-200	------	2	
CB	200X800	3T16	------	3T16+3T16			3T12	T10-150	------	T10-200	------	2	
CB1	200X1200	2T16	------	3T20+3T20			4T12	T10-150	------	T10-200	------	2	

Govemment Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

حكومة رأس الخيمة
دائرة البلدية
إدارة المباني

Approved On — اعتمد في

15 November 2024

Building Permit #: PRMT-24-0005227

Appoved By:

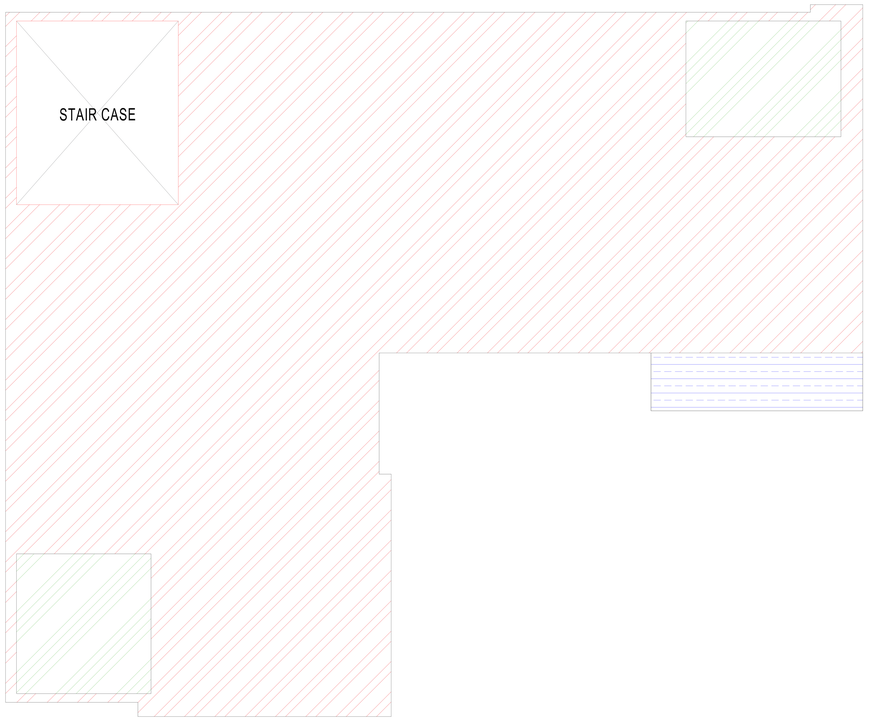
Savana
Engineering Consultancy & Interior design

Client	SAEED AHMED MOHAMMED HASAN ALMANSOORI		
Project	DOUBLE STORY VILAA		
Drg.Title	FIRST FLOOR SLAB-BEAMS		
Area	ARIFAA	Floortype	
Case ID.	2673356	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	607011962
Bldg. No.	BLD-24-03157	Floor No.	
Date	OCT-2024	Sht. No.	S06
Scale	1/100	Job No.	

G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG.IBRAHIM

FIRST FLOOR SLAB-BEAMS

FIRST FLOOR



ROOF FLOOR

Govemment Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

حكومة رأس الخيمة
دائرة البلدية
إدارة المباني

Approved On — اعتمد في

15 November 2024

Building Permit #: PRMT-24-0005227

Appoved By:

ALL GLASS MUST BE DOUBLE UNITES

U-value (w/m2.k)<_ 1.8

SHADING CO- EFFICIENT (SC) <0.3

SUPER DEAD LOAD = 7.2KN/M2
LIVE LOAD = 2 KN/M2

SUPER DEAD LOAD = 6 KN/M2
LIVE LOAD = 2 KN/M2

SUPER DEAD LOAD = 3 KN/M2
LIVE LOAD = 3 KN/M2

LINEAR LOAD FOR TIE BEAMS				
TYPE	CLEAR HEIGHT (M)	THICK (M)	DENSITY (KN/M3)	LOAD (KN)
WALLS	4.30-0.24= 4.06	0.2	15	12.5
PLASTER	4.06	2*0.02	20	3.5
TOTAL SUPER DEAD LOAD				16 KN/M

PARAPET SUPER DEAD LOAD	4 KN/M

LOADING VALUE - SLAB				
LOCATION	LOAD PATTERNS		ACTUAL LOAD	TOTAL LOAD
SLAB	L.L	LIVE LOAD	2.00	2.00
	SDL	FLOOR FINISH	2.00	6.00
		SERVICE	0.50	
		PARTITIONS LOAD	3.50	
TERRACCE	L.L	LIVE LOAD	5.00	5.00
	S.D.L	DEAD LOAD	3.50	3.50
TANK (Water Height is 1M)	L.L	LIVE LOAD	10.00	10.00
	S.D.L	DEAD LOAD	2.50	2.50
STAIR	L.L	LIVE LOAD	3.00	3.00
	SDL	DEAD LOAD	4.00	4.00

SUPER DEAD LOAD = 3KN/M2
LIVE LOAD = 2 KN/M2

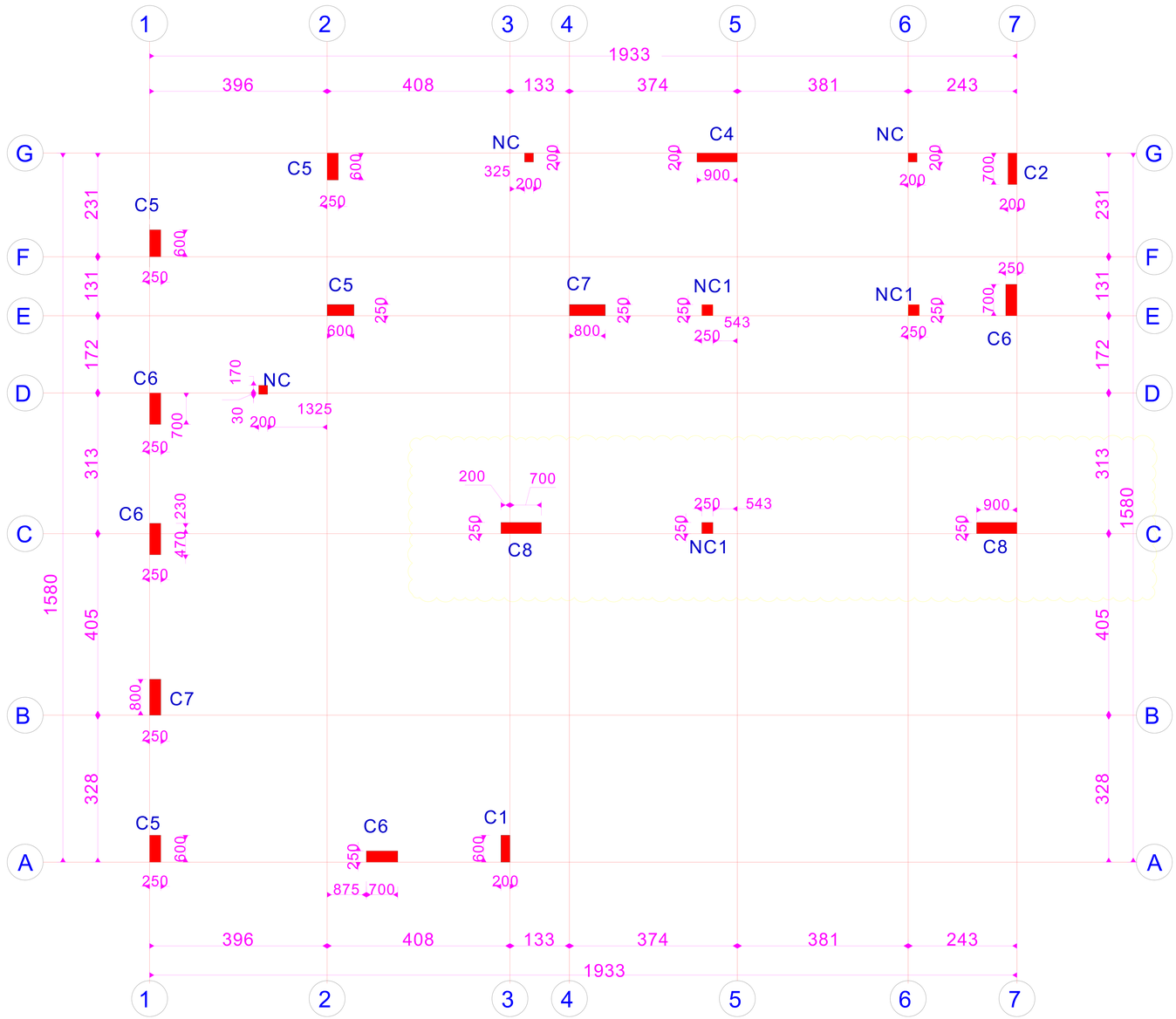
SUPER DEAD LOAD = 3KN/M2
LIVE LOAD = 10 KN/M2

FLOOR LOADING

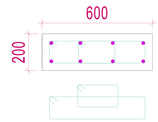
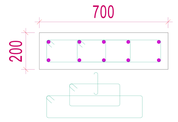
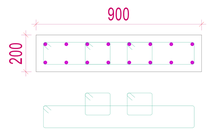
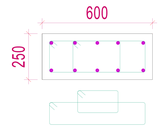
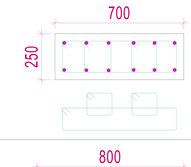
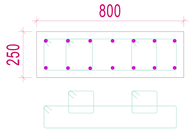
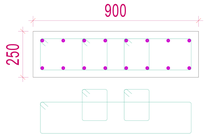
Savana
Engineering Consultancy & interior design

Client	SAEED AHMED MOHAMMED HASAN ALMANSOORI		
Project	DOUBLE STORY VILAA		
Drg.Title	FLOOR LOADING PLAN		
Area	ARIFAA	Floortype	
Case ID.	2673356	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	607011962
Bldg. No.	BLD-24-03157	Floor No.	
Date	OCT-2024	Sht. No.	S08
Scale	1/100	Job No.	

G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG.IBRAHIM



SCHEDULE OF COLUMNS

C1		Main Bar : 8T16 Links : (1T8-150)+(1T8-150) EXTERNAL INTERNAL
C2		Main Bar : 10T16 Links : (1T10-150)+(2T8-150) EXTERNAL INTERNAL
C4		Main Bar : 16T16 Links : (1T10-150)+(2T10-150) EXTERNAL INTERNAL
C5		Main Bar : 10T16 Links : (1T10-150)+(1T8-150) EXTERNAL INTERNAL
C6		Main Bar : 12T16 Links : (1T10-150)+(1T8-150) EXTERNAL INTERNAL
C7		Main Bar : 14T16 Links : (1T10-150)+(1T8-150) EXTERNAL INTERNAL
C8		Main Bar : 16T16 Links : (1T10-150)+(2T10-150) EXTERNAL INTERNAL
NC		UP TO GROUND SSL ONLY Main Bar : 4T14 Links : (1T8-150) EXTERNAL
NC1		UP TO GROUND SSL ONLY Main Bar : 4T14 Links : (1T8-150) EXTERNAL

Govemment Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

حكومة رأس الخيمة
دائرة البلدية
إدارة المباني

Approved On — اعتمد في

15 November 2024

Building Permit #: PRMT-24-0005227

Approved By:

STRUCTURAL NOTES

ACI 318-11

F'C/FCU = 28/35 N/mm2

FY = 460N/mm2

FYt = 420N/mm2

CONCRETE COVERS :

Foundations cover = 75mm

Columns cover = 40mm

Walls cover = 30mm

Beams cover = 40mm

Slabs & Stairs cover = 25mm

LEGEND:

COLUMN / WALL TO BE CONTINUED

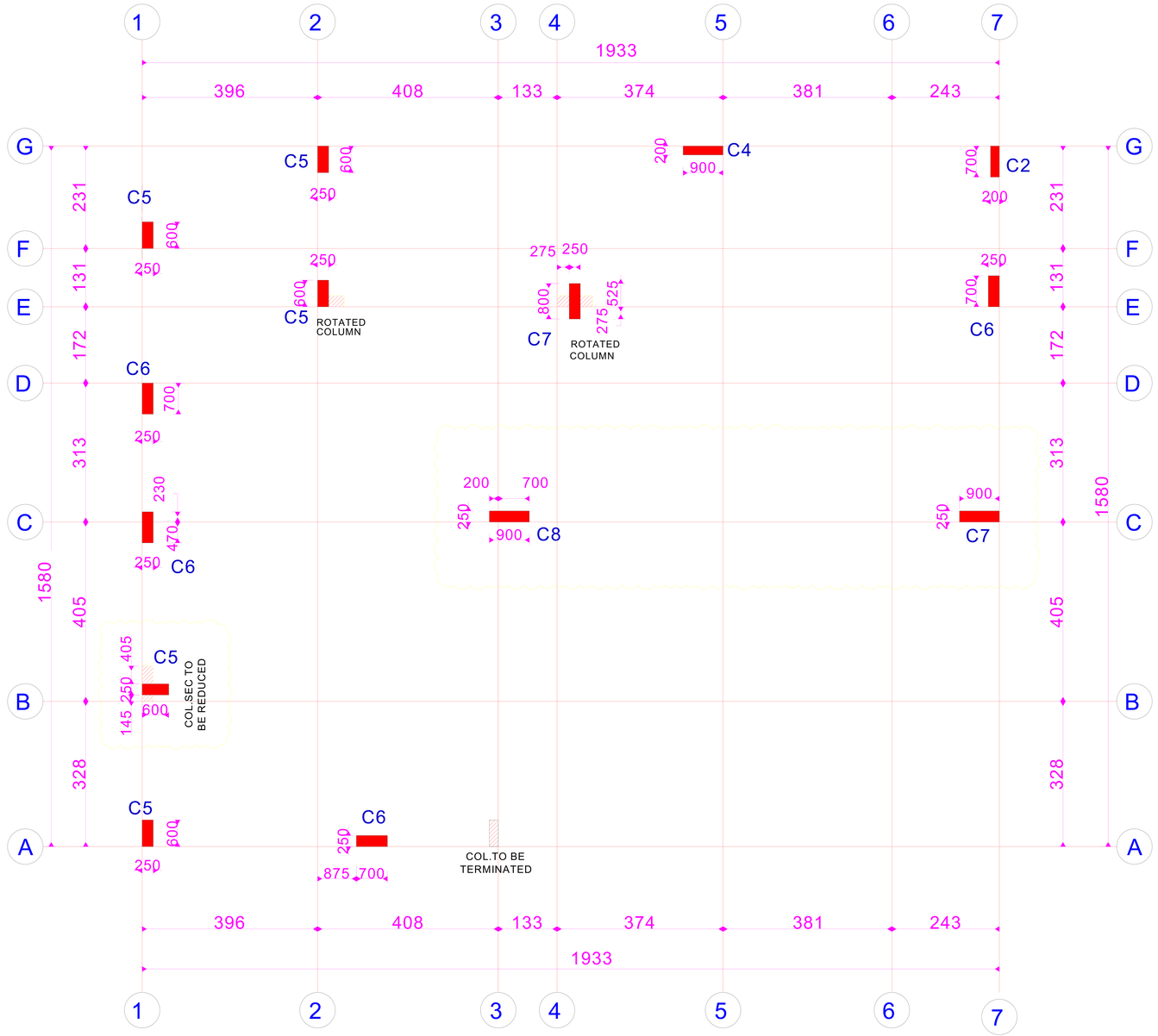
COLUMN / WALL TO BE TERMINATED

Savana
Engineering Consultancy & Interior Design

Client	SAEED AHMED MOHAMMED HASAN ALMANSOORI		
Project	DOUBLE STORY VILAA		
Drg.Title	FOUNDATION AND GROUND FLOOR COLUMN LAYOUT		
Area	ARIFAA	Floortype	
Case ID.	2673356	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	607011962
Bldg. No.	BLD-24-03157	Floor No.	
Date	OCT-2024	Sht. No.	S10
Scale	1/100	Job No.	

G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG.IBRAHIM

FOUNDATION AND GROUND FLOOR COLUMN LAYOUT



SCHEDULE OF COLUMNS

C1	600 / 200	Main Bar : 8T16 Links : (1T8-150)+(1T8-150) EXTERNAL INTERNAL
C2	700 / 200	Main Bar : 10T16 Links : (1T10-150)+(2T8-150) EXTERNAL INTERNAL
C4	900 / 200	Main Bar : 16T16 Links : (1T10-150)+(2T10-150) EXTERNAL INTERNAL
C5	600 / 250	Main Bar : 10T16 Links : (1T10-150)+(1T8-150) EXTERNAL INTERNAL
C6	700 / 250	Main Bar : 12T16 Links : (1T10-150)+(1T8-150) EXTERNAL INTERNAL
C7	800 / 250	Main Bar : 14T16 Links : (1T10-150)+(1T8-150) EXTERNAL INTERNAL
C8	900 / 250	Main Bar : 16T16 Links : (1T10-150)+(2T10-150) EXTERNAL INTERNAL
NC	200 / 200	UP TO GROUND SSL ONLY Main Bar : 4T14 Links : (1T8-150) EXTERNAL
NC1	250 / 250	UP TO GROUND SSL ONLY Main Bar : 4T14 Links : (1T8-150) EXTERNAL

Govement Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

حكومة رأس الخيمة
دائرة البلدية
إدارة المباني

Approved On — اعتمد في

15 November 2024

Building Permit #: PRMT-24-0005227

Approved By:

STRUCTURAL NOTES

ACI 318-11
F'C/FCU = 28/35 N/mm2
FY = 460N/mm2
FYt = 420N/mm2

CONCRETE COVERS :

Foundations cover = 75mm
Columns cover = 40mm
Walls cover = 30mm
Beams cover = 40mm
Slabs & Stairs cover = 25mm

LEGEND:

1) COLUMN / WALL TO BE CONTINUED
2) COLUMN / WALL TO BE TERMINATED

Savana
Engineering Consultancy & Interior Design

Client	SAEED AHMED MOHAMMED HASAN ALMANSOORI		
Project	DOUBLE STORY VILAA		
Drg.Title	FIRST FLOOR COLUMN LAYOUT		
Area	ARIFAA	Floortype	
Case ID.	2673356	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	607011962
Bldg. No.	BLD-24-03157	Floor No.	
Date	OCT-2024	Sht. No.	S11
Scale	1/100	Job No.	

G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG.IBRAHIM

FIRST FLOOR COLUMN LAYOUT